

Continuous emissions monitoring for industrial sites using low-power sensors to meet tightening regulations.

Manufacturing × Emissions and environmental monitoring × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS126 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	58 can we play, can we win	MEDIUM 0 named in the evidence	€2.9bn partial evidence · per year	NOW budgeted procurement within 1...	L1 17 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying continuous emissions monitoring for industrial sites using low-power sensors to meet tightening regulations. It targets manufacturing, chemical, and power plants that must report emissions data under new rules like the EU ETS benchmark update. The need is immediate as regulators increase scrutiny and mandate factory monitoring.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.0bn €504m – €26.0bn	€2.9bn €362m – €18.6bn	€79.8m €10.1m – €522m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€586k €586k – €1.6m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M)	30.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 3 of 4 declared geographies are outside the European reference data (BR, NG, TH); the estimate covers the European footprint only.

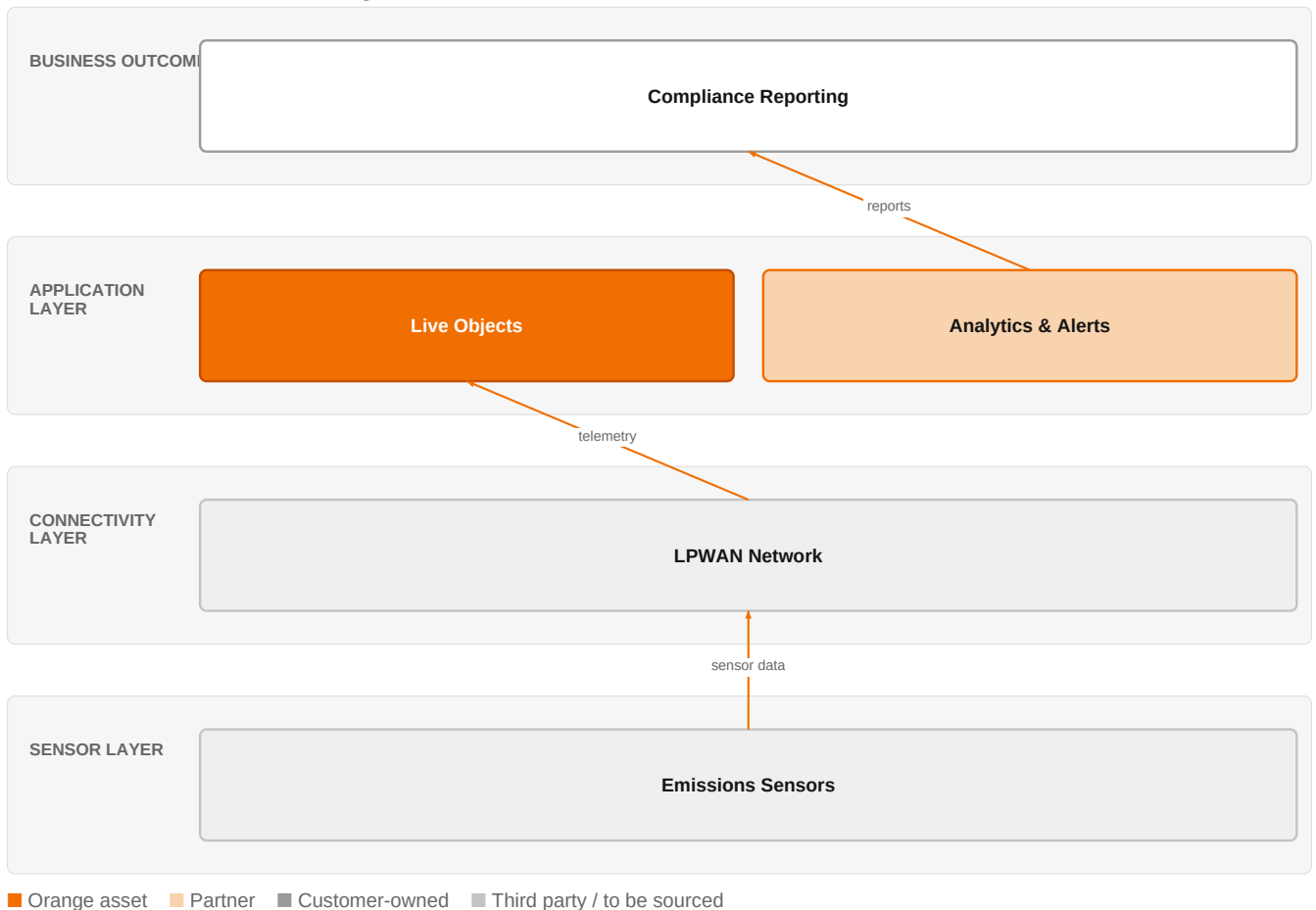
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 3 of 4 declared geographies are outside the European reference data (BR, NG, TH); the estimate covers the European footprint only.

The solution, in one picture

Continuous Emissions Monitoring Solution



This diagram shows how low-power sensors feed data through LPWAN into Orange's Live Objects platform, enabling analytics and compliance reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory pressures are intensifying: the EU ETS benchmark update puts industrial emissions data under sharper scrutiny, and Thailand has tightened air quality rules with mandatory factory emissions monitoring. Authorities are actively assessing emissions at mining sites, as seen in Nigeria. Technology is maturing with IoT-based monitoring systems and advanced anomaly detection methods for industrial sensor networks, enabling real-time alerts and reliable data analysis.

Sources: SIG-425F4DF1CB72 Envirotech Online 2026-06-20; SIG-234C749A6F8C Envirotech Online 2026-06-09; SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16; SIG-36DA075F2A6A openalex.org 2026-03-02; SIG-0E742BED215C openalex.org 2025-12-23; SIG-1C6BA6D836AF openalex.org 2025-12-17

Who buys, and why now

The buyer is typically an environmental compliance manager or plant operations director at an industrial site. The pain is the operational burden of manual emissions monitoring, which is slow and error-prone, and the risk of non-compliance with new regulations. The trigger is often a regulatory deadline, an audit finding, or a publicized enforcement action that forces them to upgrade their monitoring capabilities.

Why it is hot now — every claim bound to a dated source

- Regulatory pressures are increasing, with updates like the EU ETS benchmark putting emissions data under sharper scrutiny. [SIG-425F4DF1CB72]
- Countries are mandating factory emissions monitoring, as seen in Thailand's tightened air quality rules. [SIG-234C749A6F8C]
- Authorities are actively assessing emissions at mining sites, such as OGEPA's assessment at Ogun East Mining Site. [SIG-9E90687CF1FE]

- Regulatory pressure is increasing with the EU ETS benchmark update and Thailand's mandatory factory emissions monitoring. [SIG-234C749A6F8C, SIG-425F4DF1CB72]
- There is a demonstrated need for precise emissions monitoring technology, as shown by ABB's release of an ultra-precise system. [SIG-D48860B13392]
- EU ETS benchmark update increases scrutiny on industrial emissions data. [SIG-425F4DF1CB72]
- Thailand mandates factory emissions monitoring, indicating a global trend. [SIG-234C749A6F8C]
- ABB introduces an ultra-precise emissions monitoring system. [SIG-D48860B13392]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Live Objects platform to connect low-power sensors (LoRaWAN, NB-IoT, LTE-M) at the customer's site. The engagement would start with a site assessment, then deploy sensors, integrate data into Live Objects for real-time monitoring and alerting, and provide dashboards for compliance reporting. Orange Group researchers could contribute to advanced analytics for anomaly detection.

Why Orange can win

Orange can win by leveraging its Live Objects platform, which is a proven IoT connectivity and data management solution, and its IoT experts who can design and deploy the monitoring system. The ISO 27001 certification adds trust for handling sensitive emissions data. However, the assets are thin on vertical-specific emissions expertise, so Orange must partner with sensor vendors or analytics specialists to fill gaps.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 4.8; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators like Telia, Vodafone, and Deutsche Telekom, who offer similar LPWAN connectivity and compete on network coverage. Industrial specialists like Actility and Bosch bring deep OT domain knowledge. Global system integrators like Accenture and Capgemini can offer end-to-end integration. Sovereign cloud provider Dassault Systemes/3DS Outscale may appeal to customers with strict data sovereignty needs. Each has strengths in connectivity, domain expertise, or integration scale.

Sources: SIG-D48860B13392 Manufacturing Today In 2026-05-20

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—

Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); active in Manufacturing; competes in OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current process for emissions monitoring and reporting, and how often do you manually collect data?
- 2 Which specific pollutants or emissions are you required to report under current regulations, and have you faced any compliance issues recently?
- 3 Do you have existing sensors or IoT infrastructure on site, and if so, what protocols do they use?
- 4 What is your timeline for upgrading your monitoring capabilities, and what budget has been allocated?
- 5 Who is the primary stakeholder responsible for environmental compliance, and how do they currently receive emissions data?
- 6 Are you operating in multiple sites, and would you need a centralized monitoring solution across them?

Objections you will meet

They will say	You can answer
We already have a monitoring system in place from another vendor.	That's true, many sites have some monitoring. But with tightening regulations, you may need more precise and continuous data. Our solution can complement or replace existing systems, and we can integrate with what you have to reduce disruption.
Low-power sensors may not be accurate enough for regulatory reporting.	Accuracy is a valid concern. We work with certified sensors and can calibrate them to meet regulatory standards. Our platform also supports data validation and audit trails to ensure compliance.
We are concerned about the cost of deploying new sensors across our site.	Cost is always a factor. Our solution is designed to be scalable, and we can start with a pilot to demonstrate value. The long-term savings from avoiding fines and improving efficiency often outweigh the initial investment.

Risks and unknowns

The main risk is that regulatory timelines may slip, delaying customer urgency. There is also uncertainty about the accuracy and reliability of low-power sensors for certain pollutants, which could undermine trust. Additionally, the competitive landscape is crowded, and Orange's lack of specific emissions monitoring references may make it harder to win against specialists.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing prospect's emissions monitoring tender, using Saint-Gobain Glass and Colgate-Palmolive as references to demonstrate our ability to support industrial compliance, and highlight Live Objects' integration with LPWAN sensors.
Sales	In a meeting with a manufacturing plant's environmental compliance manager, say: 'We can help you stay ahead of tightening emissions regulations with a low-power sensor network that feeds data into our Live Objects platform, backed by ISO 27001 certification.'

Strategist / Innovator	Commission a deep-dive brief on LPWAN-based continuous emissions monitoring for manufacturing, leveraging Orange's IoT experts and Group researchers, to validate the regulatory pull and define a differentiated offer anchored on Live Objects and ISO 27001.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-BD20378972BA	2026-06-15	ted.europa.eu	1	Norway – Environmental monitoring other than for construction – Basic monitoring of environment...
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-D85076BA625E	2026-05-28	ted.europa.eu	1	Lithuania – Methane monitoring – (VPP-680) Gamtinio duju (metano) nuotekiu matavimo, duomenu an...
SIG-F0C82E295B8E	2026-03-06	ec.europa.eu	1	Pollutants and their thresholds to be reported by the biggest industrial operators under the Re...
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-0E742BED215C	2025-12-23	openalex.org	1	Improving Multivariate Time-Series Anomaly Detection in Industrial Sensor Networks Using Entrop...
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-65F732C524FB	2025-11-05	openalex.org	1	Ensuring Fish Safety Through Sustainable Aquaculture Practices
SIG-76E3106EFB64	2025-10-18	openalex.org	1	A structure-aware routing based anomaly detection for industrial multi-sensor time series
SIG-9E90687CF1FE	2026-08-16	The Guardian Nigeria News	2	OGEPA assesses gaseous emissions, pollutants at Ogun East Mining Site - The Guardian Nigeria Ne...
SIG-4D1BCBC2C864	2026-07-14	Envirotech Online	2	Mercury emissions monitoring for environmental compliance - Envirotech Online
SIG-425F4DF1CB72	2026-06-20	Envirotech Online	2	EU ETS benchmark update puts industrial emissions data under sharper scrutiny - Envirotech Onli...
SIG-234C749A6F8C	2026-06-09	Envirotech Online	2	Thailand tightens air quality rules with mandatory factory emissions monitoring - Envirotech On...
SIG-D48860B13392	2026-05-20	Manufacturing Today India	2	ABB drops ultra-precise emissions monitoring system - Manufacturing Today India
SIG-256925CCB3CE	2026-08-13	arxiv.org	3	Process-Aware Cross-Layer Adaptation for O-RAN-Enabled Industrial Systems
SIG-5C50B186142A	2026-04-30	arxiv.org	3	PhaseNet++: Phase-Aware Frequency-Domain Anomaly Detection for Industrial Control Systems via P...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:13+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.